CODEALPHA INTERNSHIP PROGRAM — BIOINFORMATICS

TASK 3: Protein Structure Prediction & 3D Visualization Analysis

Project Identifier:	CodeAlpha_Task3_Structure_Analysis
Target Biomolecule:	Human Mature Insulin (Homo sapiens)
UniProt Accession / PDB ID:	P01308 / PDB Structure: 1TRZ
Experimental Method / Resolution:	X-Ray Diffraction (1.60 Å Resolution)
Visualization & Analysis Tools:	RCSB PDB Mol* Viewer / PyMOL / UniProt3D

1. Biological Background & Structural Overview

Insulin is a vital peptide hormone produced by pancreatic beta cells that regulates glucose homeostasis. The active form of mature human insulin consists of two polypeptide chains: Chain A (21 amino acids) and Chain B (30 amino acids). Understanding its three-dimensional tertiary structure is essential for elucidating receptor binding kinetics and designing therapeutic analogs. In this task, structure PDB 1TRZ (resolved at 1.60 Å via X-ray crystallography) was retrieved and analyzed to evaluate secondary structural elements, side-chain orientations, and stabilizing disulfide bridges.

2. 3D Structure Visualization & Structural Features

2.1 Overall Secondary Structure Topology (Cartoon Representation)

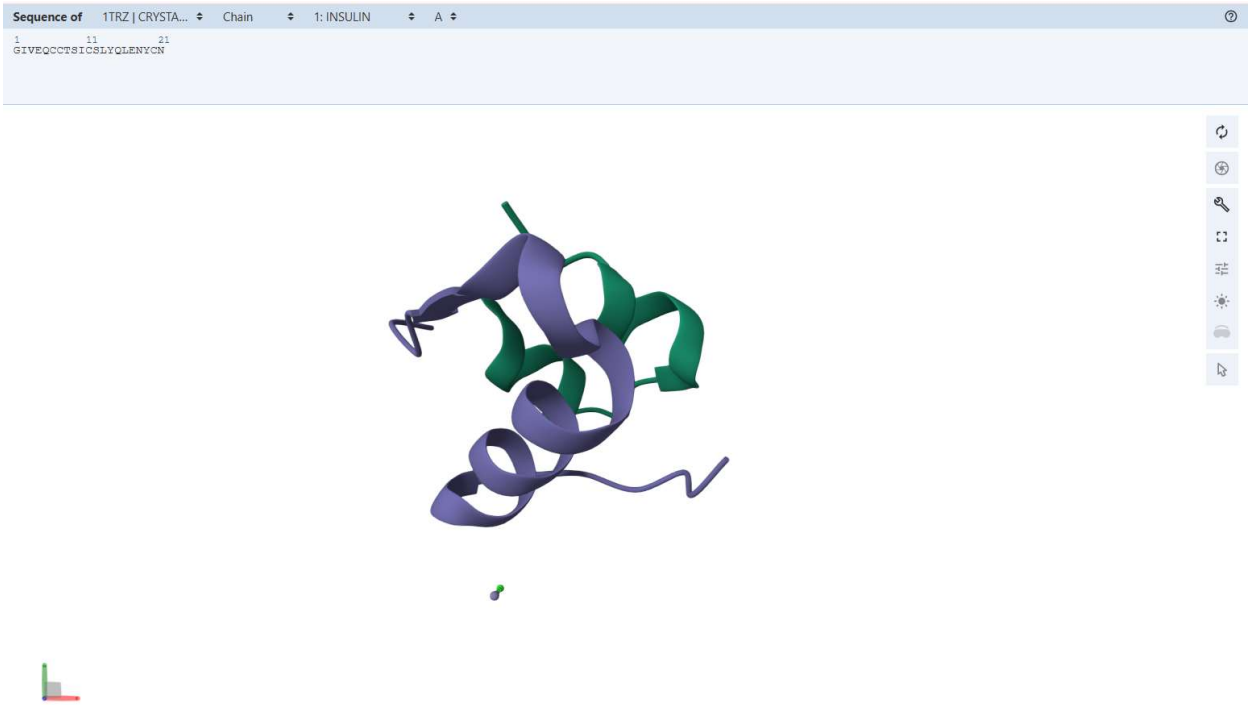


Figure 1: Secondary structure of Human Insulin (PDB: 1TRZ). Chain A (Green) and Chain B (Purple/Blue) shown in cartoon ribbon format.

2.2 Atomic Side-Chain Packing & Disulfide Bridge Coordination

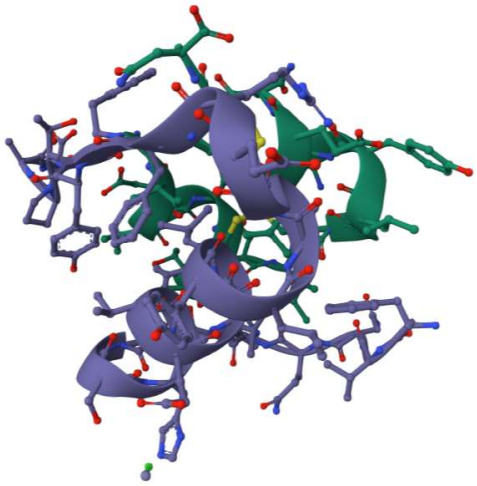


Figure 2: Ball & Stick atomistic detail of side-chains, emphasizing sulfur-containing Cysteine pairs forming covalent disulfide bonds.

2.3 Molecular Surface Envelope & Receptor Interaction Interface

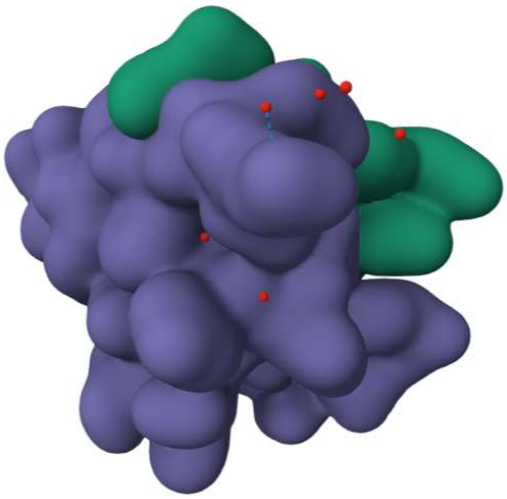


Figure 3: Solvent-accessible molecular surface representation highlighting 3D volumetric envelope and hydrophobic contact zones.

> **📌 Structural Highlights & Disulfide Linkages**
> • Alpha-Helical Content: Chain A contains two short alpha-helices (A1-A8 and A12-A20), while Chain B features a central rigid alpha-helix (B9-B19).
> • Inter-chain Disulfide Bonds: CysA7-CysB7 and CysA20-CysB19 covalently link the two chains together.
> • Intra-chain Disulfide Bond: CysA6-CysA11 forms an essential internal loop within Chain A, maintaining overall folding stability.

3. Conclusion & Integration with Sequence Analysis

The 3D structural analysis of mature insulin directly correlates with the Multiple Sequence Alignment results from Task 2. The strictly conserved Cysteine residues identified across all 7 vertebrate species are responsible for forming the three covalent disulfide bridges visualized in this task. Without these specific 3D structural constraints, the insulin molecule cannot fold into its active conformation or engage the insulin receptor (IR), demonstrating the direct relationship between evolutionary sequence conservation and tertiary structural integrity.

References / Data Sources:

1. RCSB Protein Data Bank (PDB ID: 1TRZ): https://www.rcsb.org/structure/1TRZ
2. Sehnal, D., et al. *Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures*. Nucleic Acids Research, 2021;49(W1):W431–W437.
3. PyMOL Molecular Graphics System, Version 2.0 Schrödinger, LLC.